Rubikon: Enabling Intelligent Tutoring of 3D Physical Tasks through User Modeling and On-Demand Augmented Reality

Submission #5968

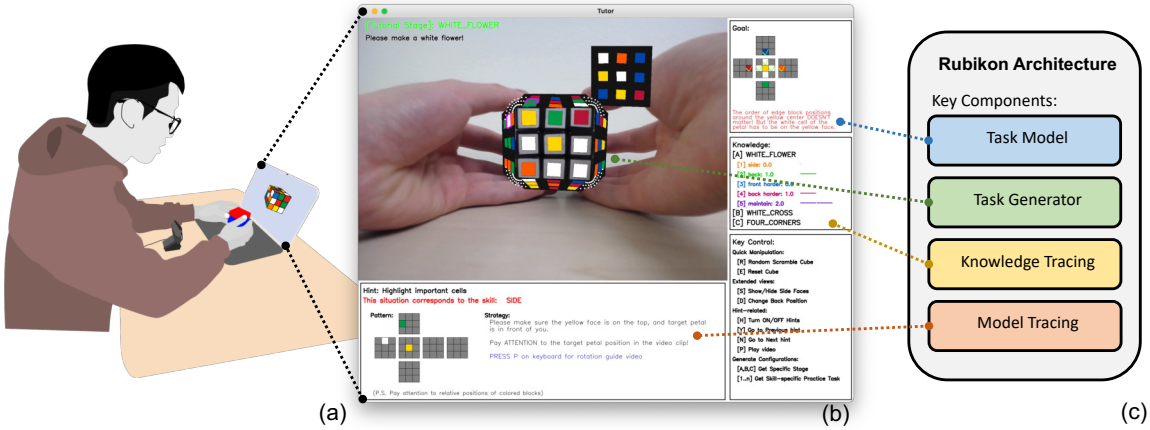

Figure 1: Rubikon is an intelligent tutoring system for Rubik's Cube learning. (a) Aruco markers are attached to each square of a Rubik's cube. The camera detects the markers and renders the color of each square in real time. (b) The user manipulates the physical Aruco marker cube while seeing the rendered cube on the screen. Rubikon provides extended views of the cube to reduce the efforts on frequent rotations. Rubikon detects the status of the cube to infer user behaviors and provides immediate feedback and hints. Rubikon automatically generates new configurations of a Rubik's Cube through AR for the user to practice unmastered skills. (c) Rubikon integrates key elements of intelligent tutoring systems which are heavily used in learning math and programming to support the learning of 3D physical tasks.

ABSTRACT

Existing video and AR tutorials for 3D physical tasks do not actively engage learners or provide adaptive feedback. Moreover, providing repeated practices for physical task learning is strenuous due to the required manual configuration of physical objects to create learning opportunities. We identify key components for 3D physical task learning that builds upon the theories of intelligent tutoring systems, with the capability to render on-demand 3D tasks through AR, and offer real-time adaptive feedback to learners. We instantiate the key components through developing Rubikon, a tutoring system for solving Rubik's Cube. We highlight three unique features of Rubikon: 1) presenting extended views of 3D objects otherwise hidden from the user; 2) generating new problems for physical tasks on-demand through AR rendering; and 3) capturing user behavior and object status through computer vision for model tracing and providing immediate feedback. We compared Rubikon with two baselines through a between-subjects experiment with 36 participants, and demonstrated the effectiveness of Rubikon in yielding higher learning gains, due to the dynamic rendering of tasks adaptive to the users' knowledge.

CCS CONCEPTS

• **Human-centered computing → Mixed / augmented reality**;
• **Applied computing → Interactive learning environments**.

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1 INTRODUCTION

Our everyday activities are full of 3D physical tasks that require a combination of motor and cognitive skills to perform, such as fixing tires, assembling a piece of furniture, and repairing a car engine. Many of these tasks are knowledge-intensive [27] and often require spatial awareness, depth perception, and hand-eye coordination during execution [6, 12]. While some tasks may only need to be performed once, others require continuous practice over time to develop mastery.

Creating tutorials to help people perform physical tasks has been a long-standing interest in the HCI community [18, 26, 27, 32, 44].

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Most prior work focuses on supporting users' performance during the execution of the tasks, e.g., through providing Augmented Reality (AR) based instructions [18, 27, 32]. While these systems have demonstrated effects on user performance on a variety of tasks, such as maintaining laser printers [17] and setting up new machines [12], they focus on supporting in-situ *performance* without the consideration to engage users in *learning* through repeated practice. Moreover, many existing tutorial systems have taken an instructor-centered perspective, where users watch videos or AR tutorials passively without practice and feedback [47]. Furthermore, for 3D physical tasks, it is often difficult for users to configure the objects to be aligned with what is taught in a tutorial video. This mismatch between reality and the tutorial may create extra hurdles for performance and learning [19].

While there has been a recent growing interest in developing tutoring systems to support the learning of physical tasks [22, 23, 45], the idea of intelligent tutoring systems has not been fully embraced in the context of 3D physical task learning. One of the key reasons is that physical task learning requires the configuration of a physical object in order for learners to practice on diverse problem scenarios, which is strenuous and hard to achieve automatically. On the other hand, intelligent tutoring systems such as cognitive tutors [4, 8, 30, 31] have achieved significant success in improving students' academic performance in subjects such as math and programming over the past decade. However, research examining their impact on students' acquisition of physical skills has been relatively scarce.

In this work, we investigate how to create AR-based tutoring systems to support learning of 3D physical tasks, through offering users deliberate practice opportunities. We identify key components in our system design aligned with several key principles in Intelligent Tutoring Systems (ITS) [30]. First, decades of learning sciences research has shown that practice and feedback make someone learn. However, in existing AR tutoring systems, users rarely get targeted feedback and repeated practice opportunities when they make a mistake. Our design supports dynamic generation of physical tasks for users to engage in repeated practice. Second, we borrow the ideas of model tracing and knowledge tracing [31, 37] from ITS and apply them in the learning of physical tasks that demand both motor and cognitive skills. Prior work of ITS mostly focused on the teaching and learning of cognitive skills, such as math and programming [30, 31]. Third, to reduce the cognitive load in learning a complex 3D physical task and to support spatial awareness and depth perception, our design visualizes hidden information that would be otherwise inaccessible to the users.

We instantiate the key components of our design through developing Rubikon, a tutoring system for solving Rubik's Cube. We use Rubik's Cube as an example application because it demands both cognitive and motor skills for execution, and has a high level of requirement for spatial awareness, depth perception, and hand-eye coordination. Rubikon demonstrates the key principles of ITS as introduced above. First, Rubikon provides learners with repeated and customized practice opportunities by generating new problems through AR rendering. This approach eliminates the need for learners to manually reset or configure the physical object to a specific state for each new problem, which can be strenuous. Second, Rubikon's intelligent tutoring backend captures users' movements and rotations and the cube's status through computer vision techniques. This allows Rubikon to provide real-time feedback to a user's moves by comparing them against a pre-defined task model (i.e., model tracing). Moreover, Rubikon uses a skillometer to track a user's weaknesses and prompts them to practice more when needed (i.e., knowledge tracing). Third, Rubikon provides extended views of the Rubik's Cube which are otherwise hidden from the user.

The foundational setup of Rubikon is the AR rendering of a Rubik's Cube, which makes the dynamic task generation and intelligent tutoring features of the system possible, as shown in Figure 3. Specifically, Aruco markers are attached to each square of a Rubik's Cube. Users can pose a camera towards the cube. The camera detects the markers and renders the color of each square in real time. The learner sees a fully-rendered Rubik's Cube through a screen while manipulating the Aruco marker cube in their hands.

To evaluate the learning benefit of Rubikon, we conducted a between-subjects experiment with 36 participants who self-identified as novices in solving Rubik's Cube. We introduced two baseline conditions to help us fully evaluate the benefit of Rubikon. Baseline 1 simulates a business-as-usual learning scenario, in which learners watch online tutorial videos to learn to solve a Rubik's Cube, while manipulating a regular cube in their hands. In Baseline 2, users are provided with an Aruco marker cube to manipulate, while seeing the rendered cube on a screen with extended views. The use of Baseline 2 is to investigate whether AR rendering of a Rubik's Cube adds to users' cognitive load when their views and hand operations are separated. In the Rubikon condition, users learn to solve Rubik's Cube with the full functionality of the system, including AR rendering, task generation, model and knowledge tracing.

We found that users who were in the Rubikon condition learned significantly more than those in the two baseline conditions. In fact, the users in the Rubikon condition scored 25% higher on the post-test, which is an authentic task that involves solving the first layer of a Rubik's Cube. This difference in scores is significant and noteworthy, as it means that the Rubikon group achieved 25% more learning in just 45 minutes of learning time. Overall, Rubikon represents an important step towards intelligent tutoring for 3D physical task learning, and the key components in our design can be extended to other domains beyond solving Rubik's Cube, such as drawing, playing the piano, or assembling furniture.

2 RELATED WORK

Rubikon builds on prior methods of Physical Task Guidance and implements innovations from the study of Intelligent Tutoring Systems (ITS). The sections following present an analysis of the existing work in these domains which informed the development of Rubikon. Much of the existing work on Physical Task Guidance concerns either the display of static information of pre-generated content, sometimes designed in an authoring interface [11, 43], or requiring real-time instructor interaction [14], only utilizing user-driven ITS for primarily non-cognitive tasks. Working implementations of ITS include features such as User and Domain Modeling and Task Selection [10] or Generation [41], but almost exclusively for traditional academic domains [16] and course settings [41], neglecting augmented reality or physical tasks [25].

2.1 Physical Task Guidance

Existing self-guided instructional methods for Physical Tasks can be separated into three different categories: Text and Figure methods, Video-based methods, and XR-based methods.

2.1.1 Text and Figure Tutorials. Requiring the least amount of technological intervention, tutorials involving only Text and Figures are the least mechanically sophisticated, cheapest, and widely used, particularly for assembly tasks [19]. Some classic applications of these methods include instruction manuals, workout diagrams, or cookbook entries. Tutorials which utilize these traditional methods, especially on paper, are considerably less expensive to create than alternative methods, but they present little opportunity for user engagement, or real-time adaptation, and identify the potential for errors in the mental conversion from conceptual space to physical space of the domain, referred to as "perception transfer" [19].

2.1.2 Video Tutorials. Video tutorials are an alternative way of teaching 3D physical tasks. Compared with text and figures, videos provide motion information in addition to static spatial relations, with demonstrations of the physical task with a sequence of movements leading to the task completion, accompanied by necessary text and audio elaborations [15]. To learn with video tutorials, users would interact with them by playing, pausing, or repeating specific portions of the video until they mastered the target knowledge.

Nevertheless, video tutorials are still far from addressing the perception transfer problem, where the users tend to get confused about how the physical object can be aligned with their perception of the object in the video [19]. Prior work has mainly been focused on augmenting the videos with easy-to-understand visual cues [35, 40]. Specifically, [40] annotates physical training videos with body highlights and visual metaphors. However, this method may not be easily generalized to more sophisticated tasks which incorporate dexterous body movements. Besides, this method builds upon the theory of imitation learning, which is difficult to apply to physical tasks that involve heavy spatial reasoning..

2.1.3 XR Tutorials. Even early publications [48] identified the incredible potential of Extended Reality (XR) in physical task training [12]. Yang et al. [48] proposed a system for a ghost-based metaphor for motion-based learning in Immersive VR, inspiring projects including ImmerTai [14], a VR system to assist learning of Tai Chi through direct guidance of an expert and AR-Arm [20], a system to master complicated arm movements using a see-though head-mounted display. These projects demonstrate the explicit benefit of XR for full-body physical learning, but lack sophisticated guidance generation.

Yamaguchi et al. [47] show that even just realtime AR annotation in the presentation of an assembly video provided reduction of cognitive load in a user study. Parong et al. [38] suggest that the adaptation of PowerPoint materials into a VR format with intermittent summarization tasks has little effect on the effectiveness of scientific communication, but contributes substantially to student engagement and enjoyability. Some AR systems exist to provide adaptive assembly guidance [11, 19, 43], but these primarily aim to support one-time task completion rather than skill development, and they neglect to include complete integration of core ITS features like personalized task generation.

XR systems to support task learning are prevalent in musical domains, such as Skrenig et al.'s AR Hero for the Guitar [42], or Zeng et al.'s FunPianoAR [49], but these too lack user modeling and task generation.

In particular, literature supports that domains which require substantial cost or effort for task construction and performance seem to experience success with XR implementation, such as the other people required in the firefighting trainer [46] and military tasks [33], or the configuration of an assembly task [19, 29, 45] and, presumably, the particular state of a Rubik's Cube.

2.2 Intelligent Tutoring Systems

The ITS paradigm is designed to be a highly adaptable collection of software tools to support individualized learning, often with the goal of reducing or eliminating teacher intervention.

2.2.1 User Model. A core component of an ITS is the ability to monitor the cognitive ability and preferences of the user, which will be referred to as User Modeling in this paper. This User Model can be responsible for several tasks within an ITS, including predicting the learning style of a user [3, 21], informing effective task generation [41], and maintaining an estimate of current user understanding [3].

Prior work has established rule-based user modeling as a method capable of producing flexible and ideal cognitive fidelity [3] and emphasized the importance of knowledge tracing throughout execution of the design [21].

2.2.2 Task Generation. Another vital feature of a successful ITS is the system's thorough understanding of the target domain [36], allowing for the meaningful interpretation of learner actions and generation or selection of impactful tasks to further user understanding.

The existing work has employed natural language processing [9, 13], machine learning and paradigmatic relation discovery techniques [41] to automatically generate tasks (or hints [10]) in terms of Multiple-Choice questions, but those are textual-based and do not extend to 3D activities. Current task generation approaches primarily follow a repetition-based principle [34], or only capture key features of the instruction to provide practice [28], lacking the ability to generate the task based on the user model, particularly for physical tasks.

2.2.3 Applying to XR. Currently, little work has been done in the application of the ITS paradigm into AR, but Herbert et al. [24] propose a set of design weaknesses of the ITS and AR systems respectively to guide future work, and later implement an example with these weaknesses in mind [25]. For AR, Herbert et al. identify overreliance on linear learning, inability to perform deep knowledge checks, and difficulties implementing scaffolded instruction; Herbert et al. find ITS' weaknesses as unintuitive instruction conveyance, difficult UI designs, and incomplete user models [24].

Herbert et al. isolate [33] as the first project to implement ITS principles into an XR application, presenting a military AR training program with simulated agents in an intricate physical environment. Subsequent projects include a Motherboard Assembly program [45] and a program that gives immediate feedback to radiology students inserting needles into tissue [5], each with demonstrated success,

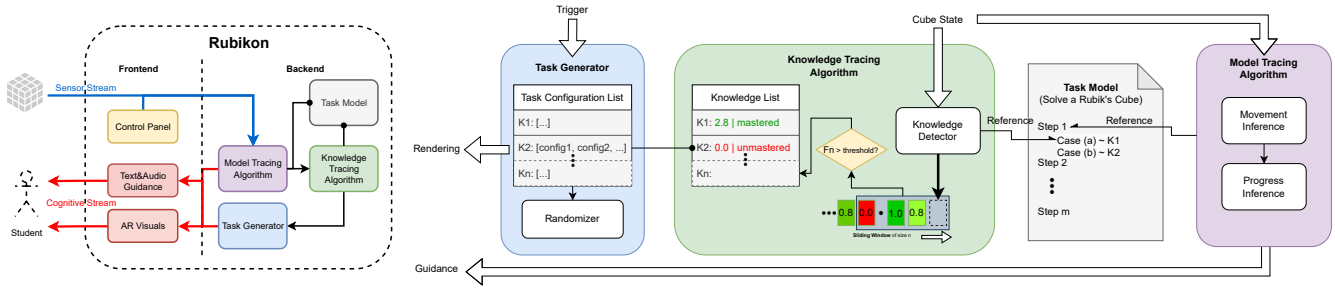


Figure 2: Rubikon architecture (left) includes a front end, which primarily provides AR rendering of the cube, and a back end (right), which consists of four key components: task model, model tracing, knowledge tracing, and task generator.

but lacking complete ITS integration and requiring considerable outside effort for reset or task setup. Herbert et al. [25] implements an ITS for network security training with an AR interface, tasking users with completing specified connections on a collection of network switches to demonstrated success, but without task generation and in a spatially simple domain.

3 RUBIKON: A 3D PHYSICAL TASK TUTOR

In this section, we first describe the key components for enabling intelligent tutoring for 3D physical task learning, then describe a prototype system, Rubikon, as an instantiation of the key components for Rubik's Cube learning.

3.1 Key Components for Physical Task ITS

To create AR-based tutoring systems to support the learning of 3D physical tasks, we aim to offer learners repeated and deliberate practice opportunities, provide targeted feedback, adapt to learners' skill proficiency, and reduce unnecessary cognitive load. Our design includes several key components inspired by ITS [30]:

(1) **A task model** that specifies the solution paths to solving the physical task. The task model determines the tutoring behavior and is used to give users hints and feedback;
(2) **A model tracing algorithm** that compares the user's actions against the task model;
(3) **A knowledge tracing algorithm** that computes the user's proficiency on the knowledge components required to solve the physical task. The user's progress is visualized through a skillometer;
(4) **A task generator** that generates tasks to target the user's weaknesses.

We instantiate these key components through developing Rubikon, a tutoring system for solving Rubik's Cube (Figure 2). We use Rubik's Cube as an example domain because it demands both cognitive and motor skills for execution, and has a high level of requirement for spatial awareness, depth perception, and hand-eye coordination.

(1) The **task model** for solving a Rubik's Cube specifies the solution paths to reach a desired state.
(2) Rubikon's intelligent tutoring backend captures users' movements and rotations and the cube's status through computer vision techniques. This allows Rubikon to provide real-time feedback to a user's moves by comparing them against a pre-defined task model (i.e., **model tracing**).
(3) Rubikon uses a skillometer to track a user's weaknesses and prompts them to practice more when needed (i.e., **knowledge tracking**).
(4) Rubikon provides learners with repeated and customized practice opportunities by generating new problems through AR rendering (i.e., **task generation**).

3.2 AR Rendering

The foundational setup of Rubikon is the AR rendering of a Rubik's Cube, which makes the dynamic task generation and intelligent tutoring features of the system possible. Specifically, Aruco markers are attached to each square of a Rubik's Cube. Users can pose a camera towards the cube. The camera detects the markers and renders the color of each square in real time. The learner sees a fully-rendered Rubik's Cube through a screen, while manipulating the Aruco marker cube in their hands. This setup enables Rubikon to automatically configure the status of a Rubik's Cube and generate tasks for users to tackle.

To enable AR rendering of a Rubik's Cube, we attach Aruco markers to each square of a cube. There are 54 (6 sides with 9 squares per side) squres in total. A fully rendered Rubik's Cube is shown in Figure 3. Before the tutoring starts, Rubikon users undergo a registration stage through which the program registers each Aruco marker on its position on the cube to construct a 3D model. This enables the program to track user rotations of the

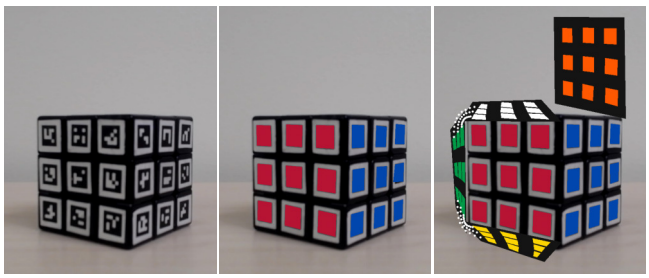


Figure 3: ArUco marker cube with rendering off (left), on (middle), and extended views (right).

cube precisely through detecting the change of markers. This AR rendering setup also enables Rubikon to automatically generate different tasks of varying difficulty for users to tackle.

3.3 Learning Objectives and Task Model

We first annotate knowledge explicitly as independent units for each stage of Rubik's Cube learning, which is inherited from the 7-stage conventional solution [1] of Rubik's Cube. Based on this annotated knowledge, we construct a task model that outlines all the knowledge required to solve the first layer as shown in Figure 4, including the steps for achieving the white flower, white cross, and four corners. One thing to note is that not all skills under one stage may appear on the cube. For example, users may only encounter the easiest skill, such as SIDE, under the white flower stage. Even if they are able to solve the white flower by themselves, they may not have fully completed learning this stage. There is no set order for them to complete the skills, but rather the goal is to master all the skills so that users can fully solve one stage in all situations. Our UI and audio guidance provide clear feedback on the user's current stage of learning and the goal they should be working towards. To help users understand the skills they need for each stage and their learning progress, we visualize them on the skillometer panel (Figure 6d) for the corresponding stage and skills.

3.3.1 Skills specified in the Task Model. In typical video tutorials, instructors often do not provide explicit information about the skills being demonstrated. Instead, they merely point to the cube and explain what to do in specific situations. This approach does not provide a comprehensive summary of the knowledge components involved. To address this issue, Rubikon specifies all skills (Figure 4) and creates the knowledge mapping for users.

3.3.2 Definition of skill difficulty level. We assign a difficulty level to each skill based on the number of times a layer or face needs to be switched during rotations. For example, in Figure 5, (a) represents the skill SIDE, which has a difficulty level of 1 star (the easiest) in Figure 4 because users only need to rotate the up layer counterclockwise to align the target squares. On the other hand, (e) represents the skill BACK HARDER, which requires users to rotate the right layer clockwise and then the up layer clockwise, making it a 2-star difficulty skill.

3.3.3 Differentiation of the skills. We match the patterns in the placement of target squares for different skills on the Rubik's Cube. Take skill SIDE and skill BACK HARDER for instance. In Figure 5, (a) and (c) may appear different but belong to the same skill. By expanding the 3D cube in (a) into 2D, we can see that the white square in (b) does not align with the yellow square. However, we can identify 7 similar positions (indicated by white stars in (d)) that fall under the skill SIDE. These positions are located on the sides of the faces that are closest to the yellow-center face, and have a white square in the middle. Similarly, (f) shows 4 possible positions that belong to the skill BACK HARDER in (e). These positions are located at the same horizontal or vertical level as the yellow square, on faces close to the yellow-center face, and have 3 gray squares between the white and yellow squares. By recognizing these patterns, we can better understand and specify the different skills required to solve the Rubik's Cube.

Task Model (for first layer of the cube)

1. White Flower Stage	DIFFICULTY
a. skill_1: SIDE	*
b. skill_2: BACK	*
c. skill_3: FRONT Harder	**
d. skill_4: BACK Harder	**
e. skill_5: MAINTAIN	**
2. White Cross Stage	
a. skill_1: MATCH	**
3. Four Corners Stage	
a. skill_1: LEFT CORNER	*****
b. skill_2: RIGHT CORNER	*****
c. skill_3: TOP LAYER	********
d. skill_4: UNDERNEATH	********
e. skill_5: MISMATCH	********

Figure 4: Task Model: solution path to solve the first layer of the cube. Number of * means the difficulty level of the skill.

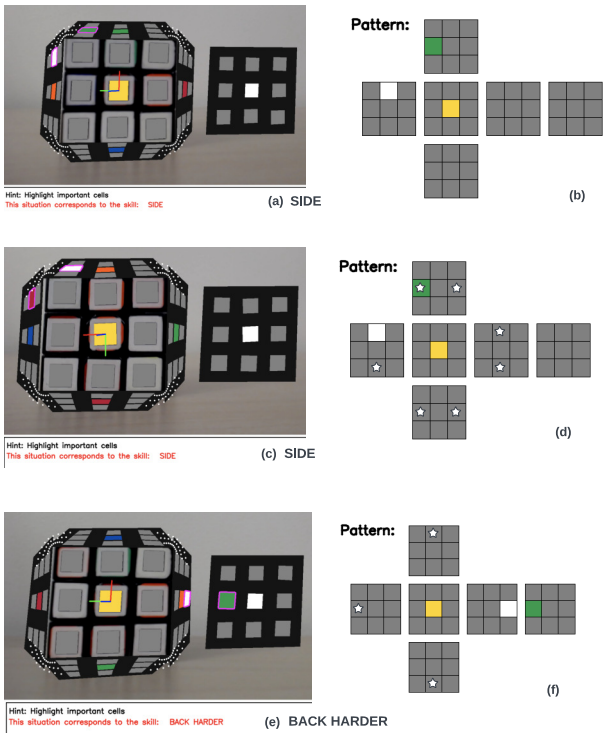


Figure 5: Rubikon Task Generation mode for specific skills based on the Task Model. Left: Cube configurations of skills SIDE (a, c), and BACK HARDER (e) rendered on the cube. Right: cube configurations corresponding the skills visualized in 2D maps in order (a-b, c-d, e-f). Different cube configurations may correspond to one skill: (a) and (c) both target at SIDE; white stars in (d) show all possible positions of the white square for the SIDE. Gray squares are unimportant.

3.4 Model Tracing and Knowledge Tracing in Rubikon

In cognitive tutors, model tracing refers to the process of tracking and analyzing students' problem-solving processes, and comparing them with a task model to give students feedback [7]. Implementing model tracing on 3D physical task tutoring is challenging since the detection of user movements is prone to errors. Rubikon uses the status of the Rubik's Cube to infer the learner's behavior, which is convenient and accurate. When a learner uses Rubikon to learn, the knowledge tracing algorithm detects and interprets the cube state transition sequences. The state transition sequences are a list of block positions. A state is pushed to the list when the user makes a move. When a block is detected to be in-place, the system gives real-time positive sound feedback to the user (unless the user makes too many useless moves before this state). When the user struggles, they can request hints. Rubikon provides multi-level hints to the user as in traditional cognitive tutors.

In addition to model tracing, Rubikon parses the sequences of state transitions to detect the knowledge component the user just exercised for the purpose of knowledge tracing. The knowledge tracing algorithm keeps track of the user's success and failure with respect to each knowledge component. To detect whether an attempt A is successful (1) or not (0) from its sequence of state transitions $S\{k\}$ which has infinite permutation, we simplified the process by first creating an ordering function $Ord(\cdot)$ describing the complexity (how difficult it is) of each state ("Difficulty" in Figure 4), then calculating the ratio of transitions where the order of the new state is smaller than that of the previous state, and comparing it with the threshold t_1 (default 0.8):

$$Ratio = \frac{\sum_{i=1}^{k-1} [\![Ord(S_i) < Ord(S_{i+1})]\!]}{k}, \tag{1}$$

$$A = [\![Ratio > t_1]\!]. \tag{2}$$

The system then uses a scoring function $S(\cdot, \cdot)$ to compute the user's score on each knowledge component within the most recent n attempts $A\{n\}$ (default 3). The score is adjusted if the user requests hints $H\{n\}$ for these attempts, based on the weight function $w(\cdot)$. The user is considered to have mastered the knowledge component if the score exceeds the threshold t_2 (default 2.4). We can formally define them with:

$$Score = S(H\{n\}, A\{n\}) = \sum_{i=1}^{n} w(H_i) \cdot A_i, \text{where} \tag{3}$$

$$w(h) = \begin{cases} 0.0, & \text{if } h \text{ is at Level 3} \\ 0.5, & \text{if } h \text{ is at Level 2} \\ 0.8, & \text{if } h \text{ is at Level 1} \\ 1.0, & \text{if } h \text{ is None} \end{cases} \tag{4}$$

$$Mastery = [\![Score > t_2]\!]. \tag{5}$$

The user's progress is visualized through a skillometer as shown in Figure 6d.

3.5 Task Generation

It is uniquely challenging for tutoring systems of 3D physical tasks to provide deliberate practice opportunities for learners, since it requires substantial efforts to reset or configure the physical objects to a certain status. Rubikon enables automatic task generation for 3D physical tasks through AR rendering, as introduced in 3.2. In particular, Rubikon generates tasks that target the knowledge components the user has not demonstrably mastered. For example, if the user has not mastered solving petals[1] in the BACK Harder position, Rubikon generates configurations of the Rubik's Cube that specifically targets this knowledge component, so that the user can engage in repeated practice.

There are two mechanisms that would trigger the task generator. First, Rubikon automatically generates tasks after a user completes the tutorial stage but has not demonstrated mastery of some knowledge components. Rubikon prompts the user to further exercise such knowledge components and updates the skillometer. Second, users can request the system to generate practice tasks for them if they want to exercise a specific knowledge component further.

3.6 Rubikon User Interface

The Rubikon user interface contains 5 sections, as shown in Figure 6.

(1) **The Playground.** The top-left section displays the rendered Rubik's Cube along with the on-demand AR capabilities.
(2) **Hint Panel.** The bottom-left section describes the hints, i.e., what the user should do to achieve the goal. The Hint Panel will display varying information depending on the level of the hint.
(3) **Goal Image.** The top-right section provides a 2D view of the desired cube state to achieve in the current stage. Only the relevant squares are shown, while the rest of the squares are grayed out.
(4) **Skillometer.** The middle-right section visualizes the skillometer based on the knowledge tracing algorithm.
(5) **Controls.** The bottom-right section lists the key-press events users can trigger. This includes resetting and scrambling the cube, turning on and off extended views, requesting hints, and generating tasks.

Rubikon provides two categories of on-demand AR functionalities to support learning.

3.6.1 Extended Views. We create a multi-angular view of the Rubik's Cube by showing its hidden sides. Through an iterative design process, we eventually set the default configuration to be 4 trapezoidal shapes for the left/right/up/down sides and a shrunk mirror image for the back side (Figure 3). While the default location of the back view is set to be top-right, users are able to change it to match the sides views (Figure 7). Squares of the same petal of the cube separated into two sides are connected with an arc-shaped line of dots, as an indicator of spatial connectivity. The extended views are geometrically computed and adapt to different orientations of the cube. The goal of the extended views is to reduce the cognitive load on learners when they process complex spatial information while at the same time remembering the formulas to solve a Rubik's Cube.

3.6.2 Multilevel Hints. To provide users with necessary help when they struggle in the learning process, Rubikon provides three levels of hints (Figure 8):

[1] Rubik's Cube blocks with two squares, also referred to as "edge blocks."

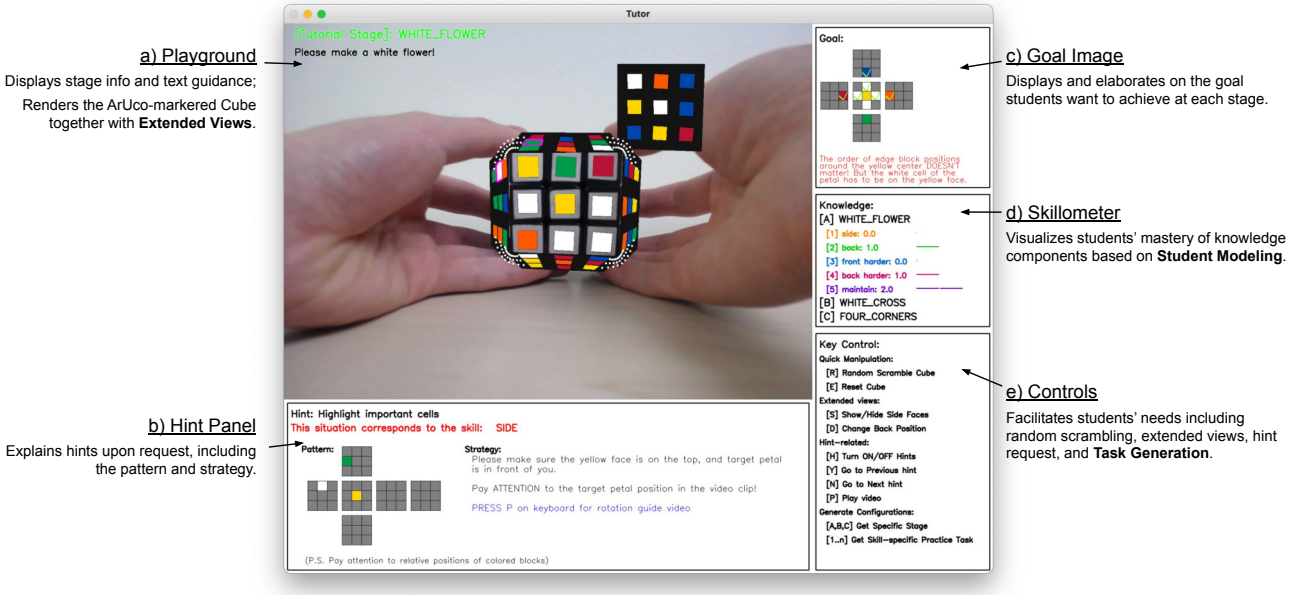


Figure 6: Rubikon User Interface, consisting of a) Real-time Playground, b) Hint Panel, c) Goal Image, d) Skillometer, and e) Controls Panel.

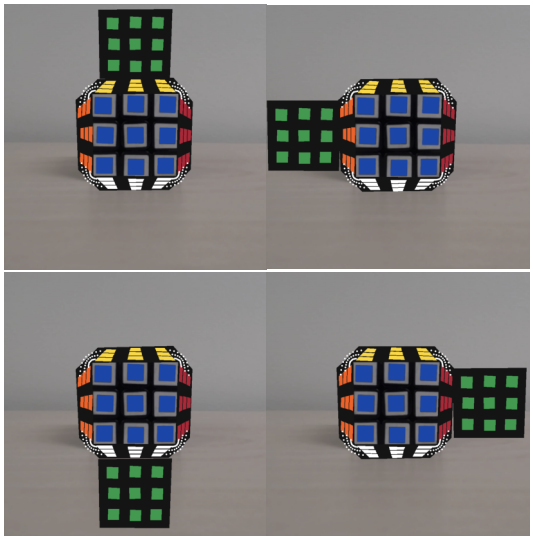


Figure 7: Extended views with alternative back side locations.

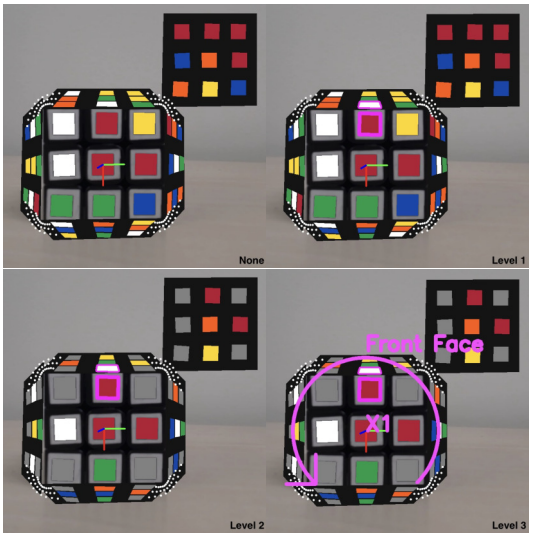


Figure 8: Rubikon views at different hint levels.

(1) **Level 1:** Highlight the target block and destination.
(2) **Level 2:** Gray out unimportant blocks.
(3) **Level 3:** Explicitly show step-by-step arrow guidance (bottom-up hints).

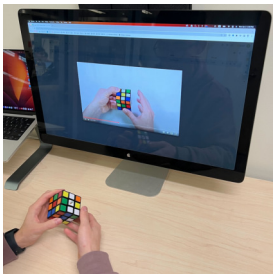
The information contained in each level of hints increases based on design principles of intelligent tutoring systems [31]. Users have full control over when to request a hint, and which level of hints to use. Requesting a hint will lower the user's score on the corresponding knowledge component, which indicates that the user needs more practice on that knowledge component.

4 EVALUATION STUDY

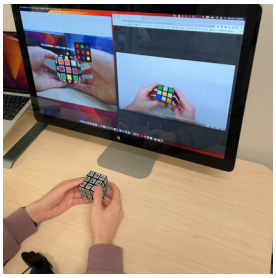
We performed an IRB-approved evaluation study to understand the learning benefits of Rubikon in comparison with business-as-usual video tutorials. One potential risk of the AR-rendering of the Rubik's cube with extended views is that it may add to the user's cognitive load since the user's views and hand operations are separated. We introduce an additional condition, in which users are provided with a rendered Rubik's Cube without the intelligent

tutoring functionalities. It allows us to investigate how users respond to an AR rendered object in comparison to the regular object. We address the following research questions:

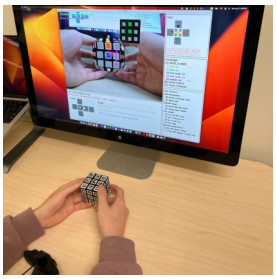
- RQ1: How usable is the AR-rendered Rubik's Cube with extended views? Does it add to or reduce the user's cognitive load?
- RQ2: When users learn to solve a Rubik's Cube with the three methods, namely, business-as-usual video tutorials with a regular Rubik's Cube, video tutorials with a rendered Rubik's Cube, Rubikon system, which leads to higher learning gains?
- RQ3: What are users' experiences when they use the Rubikon system? What challenges do they encounter and what are the design implications for develpoing intelligent tutoring systems for learning 3D physical tasks?



(a) Condition (1): users used a regular cube and followed the video tutorial.



(b) Condition (2): the left side on the screen is Rubikon rendering the color and extended view, and the right side is the video tutorial.



(c) Condition (3): Full Rubikon system.

Figure 9: Study Design. Participants are assigned to one of the conditions.

4.1 Study Design

To address the research questions above, we designed a between-subjects experiment in which participants learned to solve a Rubik's Cube in one of three conditions, as shown in Figure 9. We adopted a business-as-usual baseline in which participants learned to solve a Rubik's Cube by watching online video tutorials, while manipulating a regular cube (Condition 1). In Condition 2, the users manipulate a render cube with extended views. The user is able to adjust the position of the back side in the AR view. At the same time, users watch the same online video tutorials as the users in Condition 1. In Condition 3, users experience the full functionality of Rubikon, with both AR rendering and intelligent tutoring.

4.2 Study Procedure

4.2.1 Participants. We recruited 36 participants through mailing lists at a large public university in the US. In the sign-up form, participants need to self-identify as a novice on solving a Rubik's cube. The participants had an average age of 22, with 55% Male, 42% Female, and 3% not disclosed. Participants are randomly assigned to one of three conditions.

4.2.2 Procedure. Each study session lasted 2 hours and the study was performed in person. Each participant was paid $30 in gift card. We administered a pre-test at the start of the session. Participants were presented with a goal image of a red cross and were asked to achieve this goal using a regular Rubik's Cube within 2.5 minutes. The time limit was set based on our pilot studies. Participants were assigned a score based on their performance in the pre-test. Specifically, a correct square in-place would count as one point, for a red cross, the possible total score is 8. If a participant completed the red cross within 2.5 minutes in a sophisticated way, we told them that they were not the target user of the system and asked them to leave. Otherwise, we verified that the participants were novices and proceeded with the study. After the pretest, we administered a spatial reasoning test with two tasks to measure the participant's spatial thinking ability. The spatial reasoning test contained two tasks. In task 1, the participant answered an online test, which is widely adopted to measure one's spatial reasoning ability [2]. We designed a task 2 specific to Rubik's cube, in which users are asked

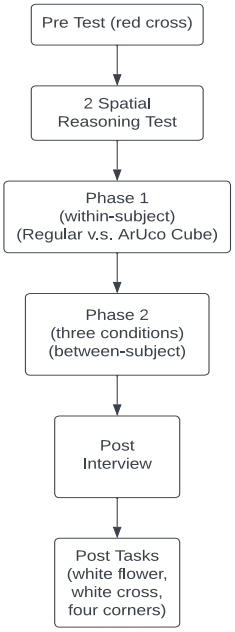


Figure 10: User study procedure.

to physically locate petals of a Rubik's cube based on a 2D map. The materials in the pre-test are included in supplementary materials.

In Phase 1 of the study, we aimed to investigate whether users were able to successfully manipulate an AR rendered Rubik's cube, and whether the AR rendered cube with extended views added to the user's cognitive load. Users were asked perform a series of 5 tasks to move a specific square to a targeted location with either a regular cube or an ArUco markers cube. Users were counterbalanced on which cube to manipulate first. We recorded the time it took each user to perform each task. Participants answered the NASA Task Load Index (TLX) survey [39] after they manipulated each cube. This allowed us to assess the cognitive load the users had while completing the tasks. We then conducted a short interview with the participants on their experiences manipulating a regular cube versus an AR rendered cube. We asked questions about whether they found the extended views in AR to be helpful or distracting for them to locate specific squares and petals. Participants in all three conditions experienced the same procedure for phase 1.

In Phase 2, participants were randomly assigned to one of the three conditions and were given 45 minutes to learn to solve the first layer of a Rubik's Cube. In conditions 1 and 2, users could watch the tutorial video freely. We selected a popular online video tutorial that breaks down the task into three stages and addresses all the situations in solving the first layer of a Rubik's Cube. In condition 3, users learnt with Rubikon, in which they worked to achieve a goal in each stage. When they struggled, they could request a hint. Throughout the learning process, they could request to get more practice on certain knowledge components.

Following phase 2, the participants answered the NASA TLX questionnaire [39] again. We then performed a short interview and probed into participants' learning experiences. For participants in the Rubikon condition, we specifically asked questions about their experiences with the adaptive task generation functionality, and their thoughts on the accuracy of the model tracing and knowledge tracing algorithms.

In the end, all participants did a post-test, which is to solve the first layer of a Rubik's Cube. We consider this to be an authentic post-test that reflects the system's tutoring objectives. We broke down the post-test into three tasks, in alignment with the three learning objectives, namely solving "white flower", "white cross", and "four corners". The participants had 2.5 minutes to work on each task independently, before they could request a hint. We used the same scoring mechanism as we did in the pre-test, when one square is back in-place, that is one point. We applied a weighing scheme to adjust for the difficulty of the three subtasks, making the total possible point of the post-test to be 16 points. We recording each participant's post-test score and the time they took to complete the post-test.

5 FINDINGS

In this section, we report our findings in response to each research question. Before we ran any analysis, we did a randomization check to make sure that there was no systematic difference in the participants' prior knowledge and ability across the three conditions. We ran a one-way ANOVA test on the pre-test scores scores across the three conditions. The result suggests that there is no significant difference in the pre-test scores across the three conditions ($F(2,33) = 0.11$, $p = 0.90$).

5.1 Changes in Cognitive Load When Using the AR Setup

Our first research question aims to answer whether the AR-rendered cube with extended views is usable and whether the user is able to accurately manipulate the rendered cube in a similar way as a regular Rubik's cube. In our study design, we answer this question in both phase 1 and phase 2.

In phase 1, we did a within-subjects comparison where participants completed a petal identification task using either an AR rendered cube or a regular cube. Each participant did 5 tasks with an AR rendered cube, and 5 tasks with a regular cube, the order of the two cubes is counterbalanced. We recorded the time it took the user to locate the petal. We then ran a paired-samples t-test using the time taken to complete each task. The result shows that it takes longer for the user to complete the tasks with a rendered cube ($t(179) = -5.71$, $p<0.01$). It is not surprising since the rendered cube uses new technology and requires higher degree of eye-hand coordination. However, we did notice that the time taken to complete a petal identification task decreased over time as shown in Figure 11.

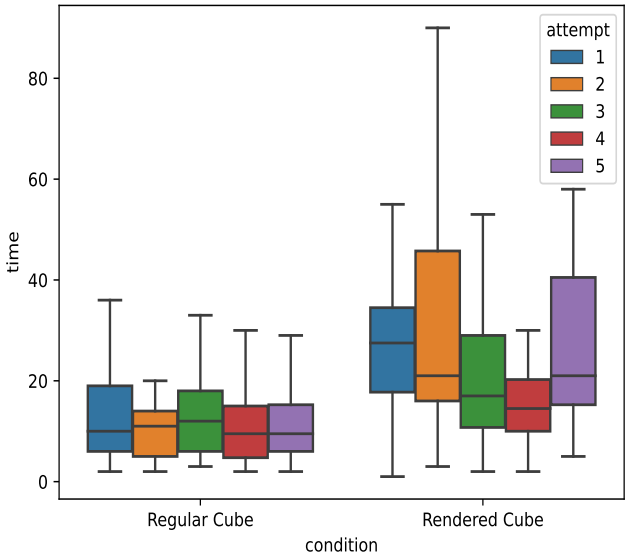


Figure 11: Phase 1 completion time for regular cube and ArUco cube.

In phase 2, the comparison between conditions 1 (regular cube with video tutorial) and 2 (rendered cube with video tutorial) allows us to further compare the cognitive load incurred by the AR rendered cube over a long period of time. The NASA TLX result in phase 2 suggests that the cognitive load the users reported in Condition 1 and 2 is comparable, as shown in Figure 12.

AR rendering of a Rubik's Cube is a foundational step in Rubikon, which makes it possible to automatically generate new configurations of a physical object and provide repeated and targeted practice opportunities for the students. As a summary, we see that although

it takes longer for participants to locate squares at the beginning, the delay was reduced over time. Once participants are familiar with the setup, they reported a similar level of cognitive load in phase 2 of the study.

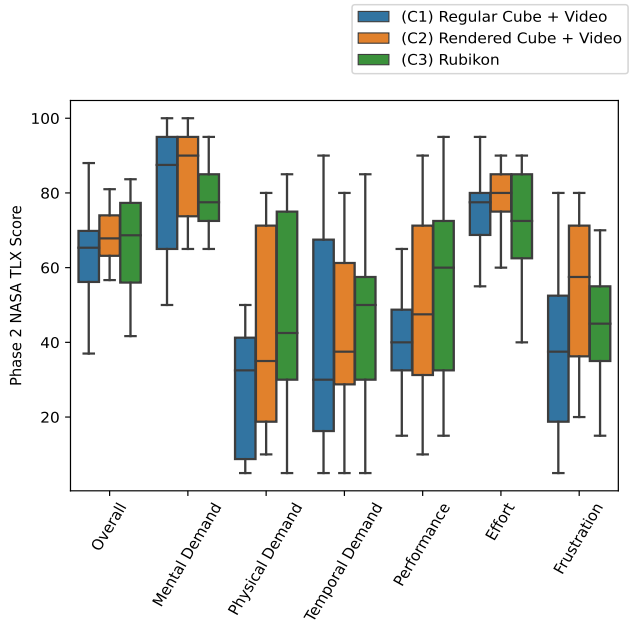


Figure 12: NASA TLX Score of Phase 2 where users learned to solve the first layer of the cube through different conditions.

5.2 Rubikon Leads to Higher Learning Gains

In order to investigate the learning benefit of Rubikon, we built a linear regression model, with the post-test score as the dependent variable, and the condition as the fixed effect, we also included the learners' pre-test score, and the time they took on completing the post-test as covariates. We found that users in Rubikon had significant higher learning gains compared to the two baselines, with the average scores shown in Figure 13. With the pre-test score and post-test task completion time controlled, the regression model suggests that, when a user is assigned to the Rendered Cube condition in contrast to the Rubikon condition, they will on average achieve 2.9 points lower in the post-test ($t=2.43$, $p = 0.02$); when a user is assigned to the Regular Cube condition, they will on average achieve 3.8 points lower in the post-test ($t=3.25$, $p=0.002$). This is a considerable effect size as learners achieved 3-4 points higher (25% of the whole task) through Rubikon in 45 minutes of the learning time.

5.3 User Experiences and Design Implications

We transcribed the interview portions during the user study and analyzed the notes using affinity diagram (See A.1).

5.3.1 Mental & Physical Load. Different participants experienced different mental and physical loads. To some participants, the AR rendering of Rubikon constrained the way they manipulated the

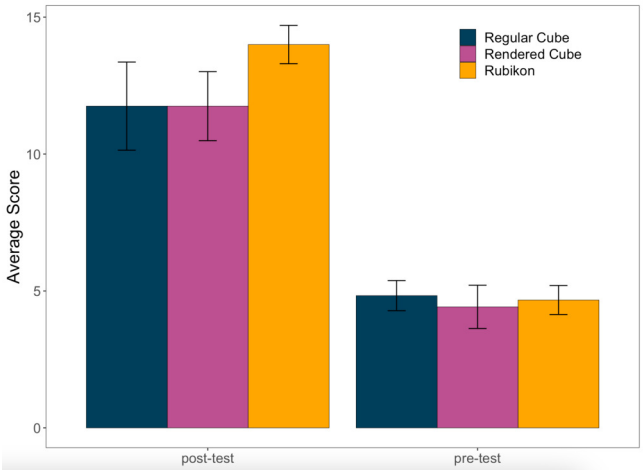


Figure 13: Users in the Rubikon condition had statistically significant higher learning gains compared to the users in the two baseline conditions.

cube, which resulted in higher physical efforts (P11, P14, P25, P7), mental efforts (P5, P21, P22, P28, P35), and thus higher cognitive load. Some participants pointed out that Rubikon required more spatial thinking compared to a regular Rubik's Cube (P5, P17, P19, P20, P24).

However, this additional cognitive load tended to decrease over time as shown in Figure 11. As some participants (P10, P11, P12, P14, P18, P33) pointed out, after some time, they became used to the Rubikon system. P10 commented, *"Rubikon was new so it took some time getting used to. It took some time before I could use it as well as the normal Rubik's Cube."*

5.3.2 Extended Views. Rubikon creates a different visual system than that of using regular Rubik's Cubes with extended views showing hidden sides. Multiple participants (P13, P14, P18, P20, P25, P26) shared that they were able to use the extended views to reconstruct the same 3D space as when using a regular cube, specifically with the visual aids Rubikon provided. *"I think I got the point where the virtual cube has the same spatial awareness as the physical cube. (P14)"* Many participants (P7, P11, P6, P25, P18, P13, P27) also consistently mentioned that the extended views provide just enough amount of information that was easily understandable, *"Helpful to present more information, not too overwhelming, straightforward. (P7)"*. With the extended views, participants could quickly locate the squares they were looking for (P17, P10, P2, P34, P21, P26, P31), without the need to physically manipulate the cube to another side (P4, P6, P25, P32, P35).

Participants also shared concerns about the usability of the extended views. P35 and P22 mentioned that the squares in the extended views were too small, and P29 said that "it was hard for me to visualize the blocks on the back side."

5.3.3 User Modeling. Participants found the user modeling functionalities of Rubikon to be accurate. Several participants (P4, P7, P19, P28) pointed out that the knowledge tracing algorithm was able to track the knowledge components they did not master well.

"I felt like I did struggle with those low scores skills. (P4)" Participants (P13, P34) also found the skillometer of Rubikon to be useful. P34 mentioned, *"I think it was useful to know which skills I still need practice with so I could pay more attention during the tutorial."*

5.3.4 Task Generation. With knowledge tracing, Rubikon generates tasks that target the knowledge components that the user hasn't mastered. This is a key component in the design of Rubikon. In most prior tutoring systems for physical tasks, repeated practice opportunities of different scenarios are limited, since the physical object usually contains rigid parts, and it takes efforts to configure a physical object to a certain status. Because of this, tutoring systems often provide linear step-by-step guidance for users. The AR setup of Rubikon enables it to generate new tasks for a user to practice a certain skill. Throughout the user studies we observed cases that users in the baseline conditions didn't exercise the important skills to solve a Rubik's Cube.

In the baseline conditions, multiple participants (P17, P27, P35, P33, P11) mentioned that they were at times frustrated by the mismatch between the configuration of the cubes in the video and on their hands. "When mismatches occur, I don't know how I can get there. (P27)" Participants (P9, P15, P14, P17, P18, P35) also pointed out that the tutorial videos went over the process of solving a Rubik's Cube linearly and chronologically. Many participants either wasted time watching the portions that they had already mastered (P14, P17, P18, P29, P33), or failed to keep up with the pace of the video (P9, P12, P14, P15, P17, P18, P22), leading to an incomplete understanding of the content taught in the video (P9, P22, P29, P33).

In contrast, participants in the Rubikon group did not encounter such problems. First, participants liked it that Rubikon automatically generates problems targeting each knowledge component (P1, P25). "I liked how it taught specific skills and provided videos with algorithms. (P25)" Second, participants also appreciated the extra repeated practice opportunities on the skill that they hadn't mastered (P1, P10, P32, P13, P34). *"I was able to learn more as the practice mode gave the practices that use the same concept for multiple times. (P34)"*

6 DISCUSSION AND FUTURE WORK

While we acknowledge that using the Rubikon system may be slightly more challenging than using a regular cube due to its additional features and hidden information, it is still a valuable tool for learning to solve a Rubik's Cube and improving spatial reasoning skills. The insights we gained could inform future work for designing ITS for 3D physical tasks.

6.1 Tradeoff between cognitive load and consolidation of learning

From the data collected on the usability of the system during the user study session, participants indicated that the Rubikon system was more difficult to manipulate than a regular cube. However, we observed that while manipulating the ArUco cube, participants spent more time strategizing and considering how the colors would change position ahead of rotation in order to optimize their solution and avoid downgrades in the user model; otherwise they would get stuck in the practice loop of their weak skills. This enhanced their memorization of unmastered skills. In contrast, when participants were presented with a regular cube and had no idea of what to do next, they tended to simply rotate the cube and wait for a new state to appear, without forming a solid strategy. This approach was not helpful for participants to develop specific skills.

To compare the Rubikon system vs. video tutorial + regular cube, we were unable to conduct a within-subject experiment due to the accumulative nature of learning. However, we asked participants in the Rubikon condition for their thoughts on the video tutorial when the problem shown was different from the state of their own cube. Some users attempted to manipulate their cube to match the tutorial problem, while others were confused or ignored the discrepancy. Although some users indicated a preference for practicing after each stage of the video tutorial, simply shuffling the cube randomly did not ensure that they would encounter the skills they had not seen before on their cube. It would require repeated attempts and a significant amount of luck for all possible situations to occur. In contrast, the Rubikon system personalized practice through easy configuration on the ArUco cube using AR, resulting in generalized improvements in users' performance. As a result, the average score of Rubikon users was consistently higher in all post-tasks compared to the other two conditions. Rubikon users also outperformed in the white corners post-task, which is widely regarded as the most challenging part of solving the first layer of the cube. In terms of time spent, Rubikon users also showed better performance compared to the other two conditions.

6.2 Generalizing to other 3D Physical Tasks

We explored Rubik's Cube learning as an example domain, while we envision that our key components for system design could be generalized to other 3D physical tasks. For example, drawing 3D objects requires strong spatial thinking abilities and the ability to imagine a perspective from which the object is being viewed. An extended view design could provide auxiliary lines or light direction to help users form a viewpoint. Additionally, the user model could track crucial skills such as shaded areas and the golden proportion, while the adaptive task generator could preset other features of the 3D objects, such as the viewpoint or object itself, leaving the specific portions that users need to practice. Another example is learning to play a musical instrument, such as the piano. In this case, the extended view could help learners read the music sheet (staff) aligned with the keys on the piano by displaying notes near the corresponding keys. User modeling could track users' rhythm, speed, coordination of playing with two hands, etc. The task generator could then produce sections of different music sheets that target users' weaknesses. In general, these design components could be helpful in improving performance and learning outcomes in other 3D physical tasks (more detail in table 1) beyond the Rubik's Cube, which we hope to explore in future work.

7 CONCLUSION

In this paper, we explored how to create AR-based tutoring systems that support deliberate practice opportunities for learning 3D physical tasks. While existing AR-based systems have demonstrated effects on user performance during task execution, they do not support repeated practice or provide feedback for skill mastery. Moreover, many systems have an instructor-centered approach,

Table 1: Brainstorming of the key components for physical task ITS design applying to other physical tasks.

Task	Hidden Info	Task Model	Knowledge/Model Tracing	Task Generation
Rubik's Cube	Extended views for 1. Sides 2. Back face	To solver the first layer, there are three stages in Task Model: white flower -> white cross -> four corners.	1. change of the cube state (caused by users' manipulation) 2. number of hint requests 3. time span 4. number of steps to accomplish	Reconfiguration of cube state to provide practices targeting users' unmastered skills (1, 2, 3, 4).
Music Instruments	1. Music notes aligned with keyboard or strings 2. Accuracy of the notes just played	To fully and successfully play a music sheet, two things need to be emphasized: 1) Accuracy of each note, and 2) Correct Beat.	1. speed 2. rhythm 3. strength 4. accuracy of each note being played	On-demand AR generating some typical sections of the music sheet corresponding to the components from the knowledge tracing (1, 2, 3, 4).
Sketch 3D objects	1. Direction of the light 2. Position of the starting view point 3. Auxiliary lines	Drawer needs 1) overall design of the drawing (what direction is the view point, what direction is the light, etc.) and 2) shaded areas in the right spot (this leads to the 3D effects)	1. consistency of golden proportion 2. spot of drawing shaded area 3. whether objects in drawing is aligned with the view point	Generate every other components from the user model but leave the one that users do not master for them to fill in (1, 2, 3).
Chemistry Experiment	1. Temperature indicator 2. Name of the devices and elements	Learner needs to be aware of 1) what kind of the experiment it would be, 2) the required equipment and elements, and 3) the rules to conduct the desired experiment.	1. Pose of manipulating various equipment 2. The sequence of conducting an experiment 3. Correct recognition of the required element	Use AR to generate the section that learners did not acquaint with (2). If learners are not familiar with the certain element, give them more practice on recognizing it (3). If pose is incorrect, learners need to redo the certain steps (1).

which creates a mismatch between reality and tutorial. To address these issues, we identified key components for AR-based tutoring systems that supports dynamic generation of physical tasks, knowledge tracing, and the visualization of hidden information to improve learning process. We instantiated these components through Rubikon, an AR-based tutoring system for solving Rubik's Cube, which provides customized practice opportunities, real-time feedback, and extended views of the cube. A between-subjects experiment with 36 novice participants demonstrated the effectiveness of Rubikon in learning outcome and the key components of our design. Overall, Rubikon represents an important step towards intelligent tutoring for 3D physical task learning, with potential applications in other domains beyond solving Rubik's Cube.

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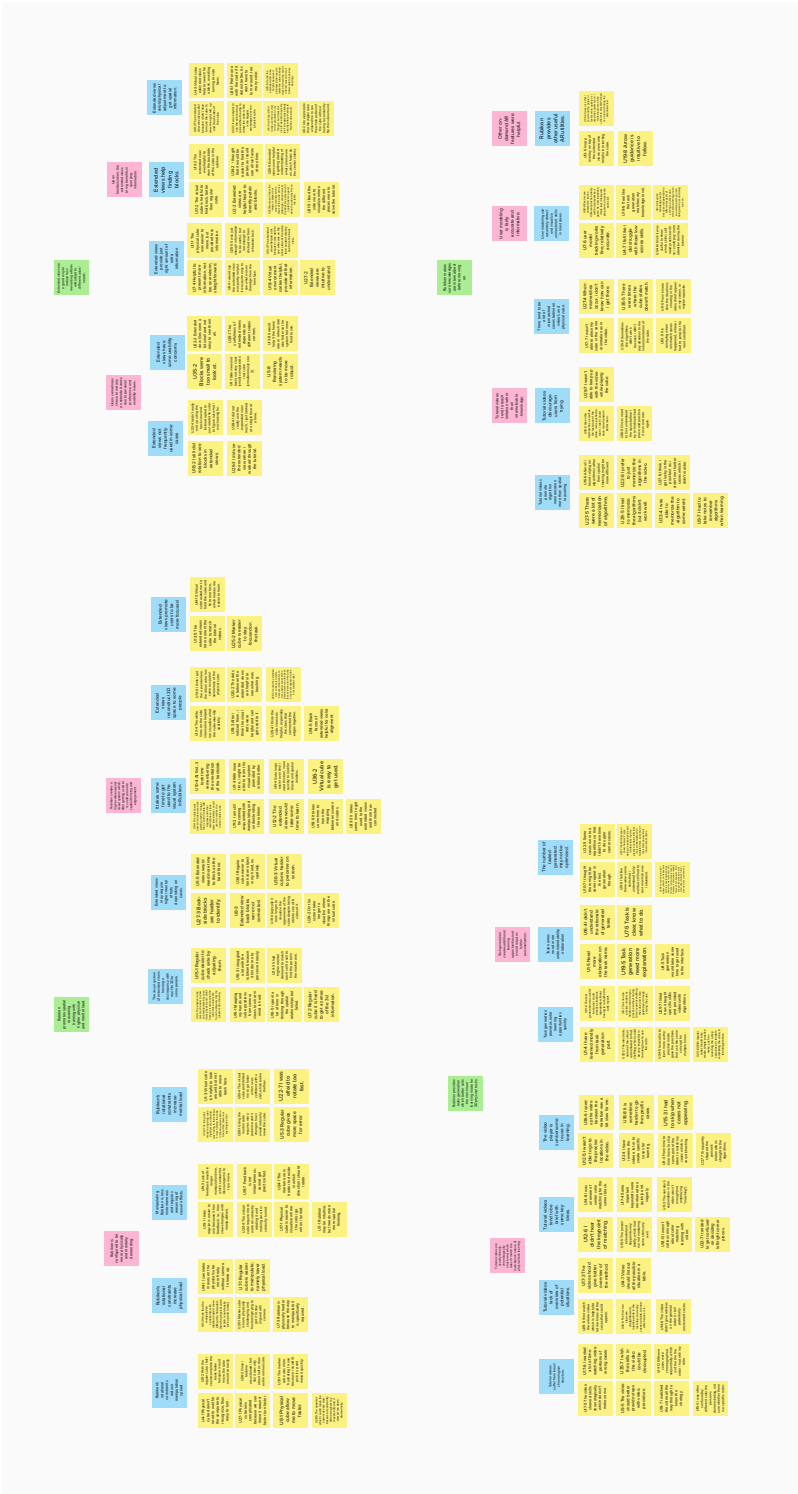
A APPENDIX

A.1 Qualitative Analysis with Affinity Diagram